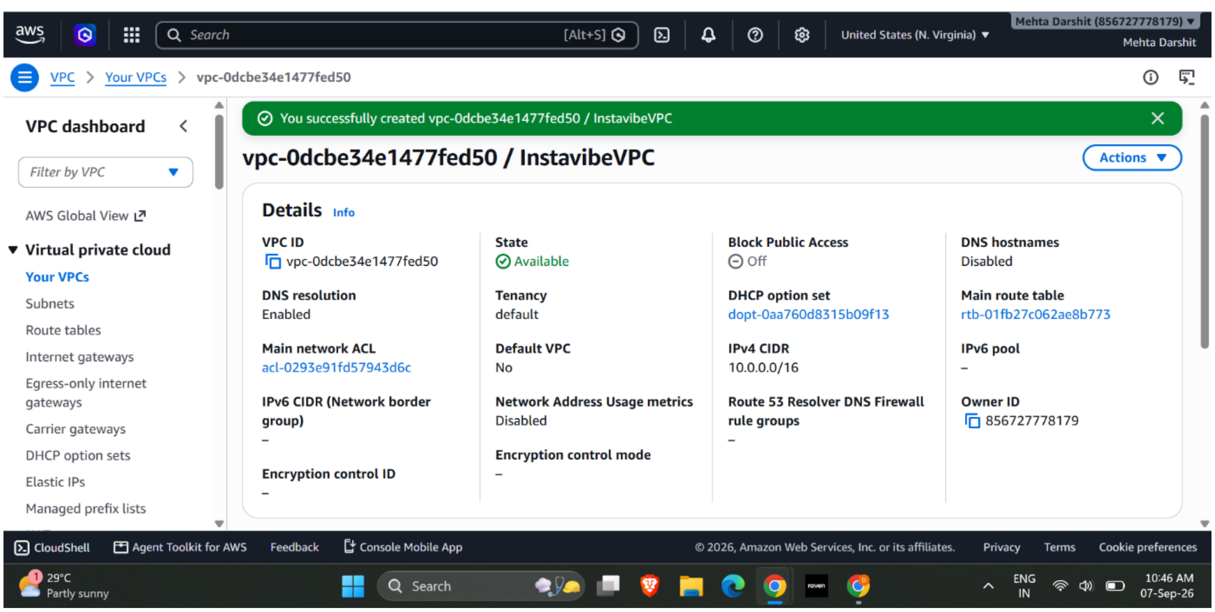
Virtual Private Cloud

Task-1.Log in to your AWS account, open the VPC dashboard, and create a new VPC named InstaVibeVPC with a custom IPv4 CIDR block (e.g., 10.0.0.0/16).

Steps

1. Log in to the **AWS Management Console**.
2. In the search bar at the top, type **VPC**.
3. Open **VPC**.
4. From the left menu, select **Your VPCs**.
5. Click **Create VPC**.
6. Under **Resources to create**, select **VPC only**.
7. Enter:
   - **Name tag:** InstaVibeVPC
   - **IPv4 CIDR:** 10.0.0.0/16
   - **IPv6 CIDR:** No IPv6 CIDR block
   - **Tenancy:** Default
8. Click **Create VPC**.
9. 

Task-2. Inside your InstaVibeVPC, create two subnets: one public subnet (e.g., 10.0.0.0/24) for hosting a web server and one private subnet (e.g., 10.0.1.0/24) for backend services.Name your subnets as 'Public-Subnet' and 'Private-Subnet' for clarity.

Create Public Subnet

1. In the VPC Dashboard, select **Subnets**.
2. Click **Create subnet**.
3. Select your VPC: InstaVibeVPC.
4. Enter:
   - **Subnet name:** Public-Subnet
   - **Availability Zone:** Choose any available AZ, such as ap-south-1a.
   - **IPv4 subnet CIDR block:** 10.0.0.0/24
5. Click **Create subnet**.

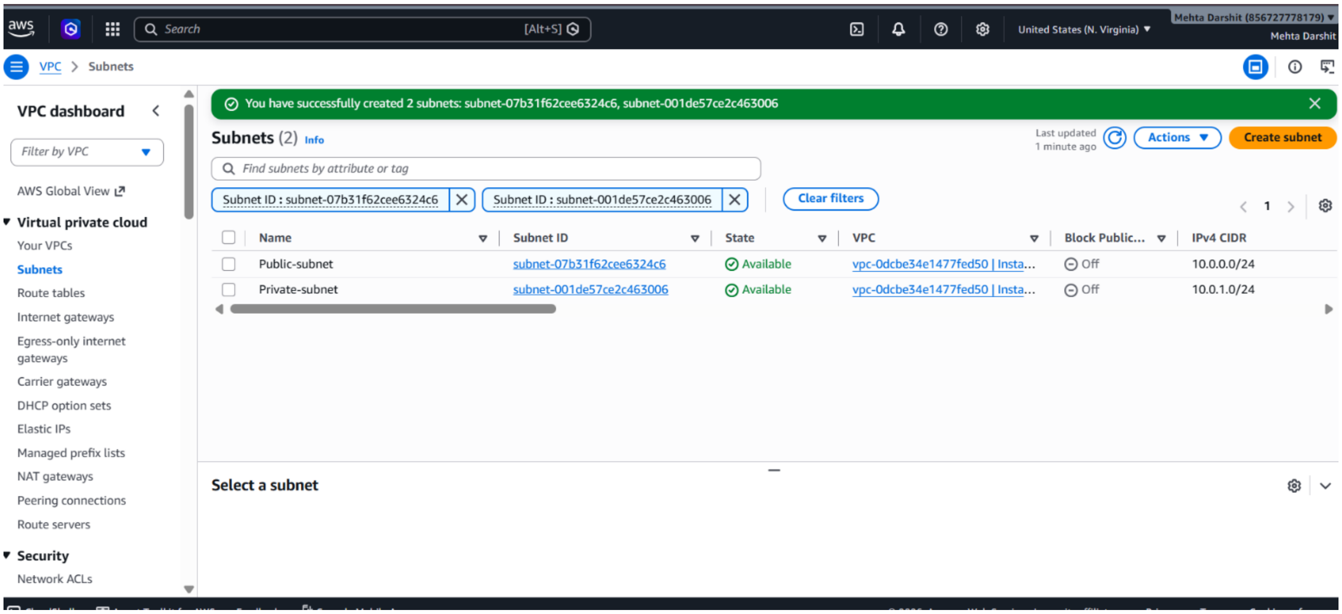
Create Private Subnet

Again select **Create subnet**.

Enter:

- **VPC:** InstaVibeVPC
- **Subnet name:** Private-Subnet
- **Availability Zone:** ap-south-1a
- **IPv4 subnet CIDR block:** 10.0.1.0/24

Click **Create subnet**.



Task-3. Configure an Internet Gateway and attach it to your InstaVibeVPC so that only the public subnet can access the internet, while the private subnet cannot.

Create Internet Gateway

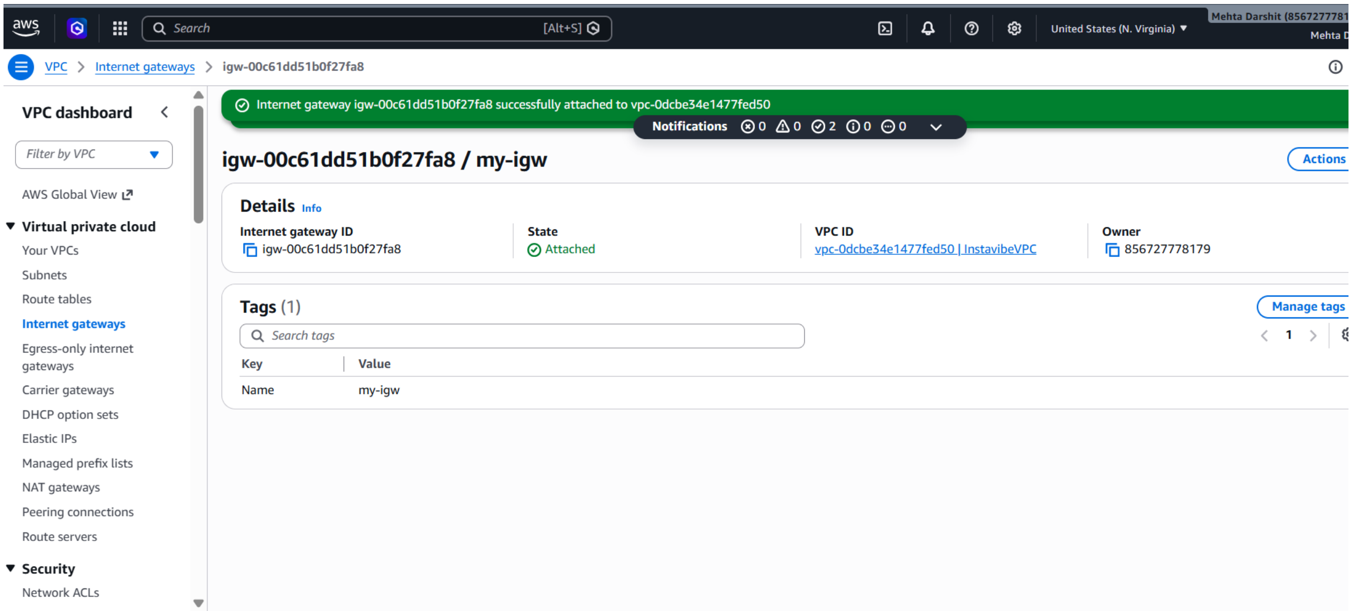
1. In the VPC Dashboard, select **Internet gateways**.
2. Click **Create internet gateway**.
3. Name it: my-igw
4. Click **Create internet gateway**.

Attach it to your VPC

1. Select my-igw
2. Click **Actions**.
3. Select **Attach to a VPC**.
4. Select:

InstaVibeVPC

5. Click **Attach internet gateway**.



Task-4 Set up route tables so that your public subnet routes traffic to the Internet Gateway, but your private subnet does not. Verify this by checking the route table associations.

A. Create Public Route Table

1. Go to **VPC → Route tables**.
2. Click **Create route table**.
3. Enter:

Name: Public-Route

VPC: InstaVibeVPC

4. Click **Create route table**.

Add Internet route

1. Select Public-Route
2. Go to **Routes**.
3. Click **Edit routes**.
4. Click **Add route**.
5. Enter:

Destination: 0.0.0.0/0

Target: Internet Gateway

6. Select my-igw
7. Save changes.

B. Associate Public Subnet

1. With Public-Route selected, go to **Subnet associations**.
2. Click **Edit subnet associations**.
3. Select:

Public-Subnet

4. Click **Save associations**.

Now the public subnet uses the Internet Gateway route.

C. Create Private Route Table

1. Go to **Route tables**.
2. Click **Create route table**.
3. Enter:

Name: Private-Route

VPC: InstaVibeVPC

4. Click **Create route table**.

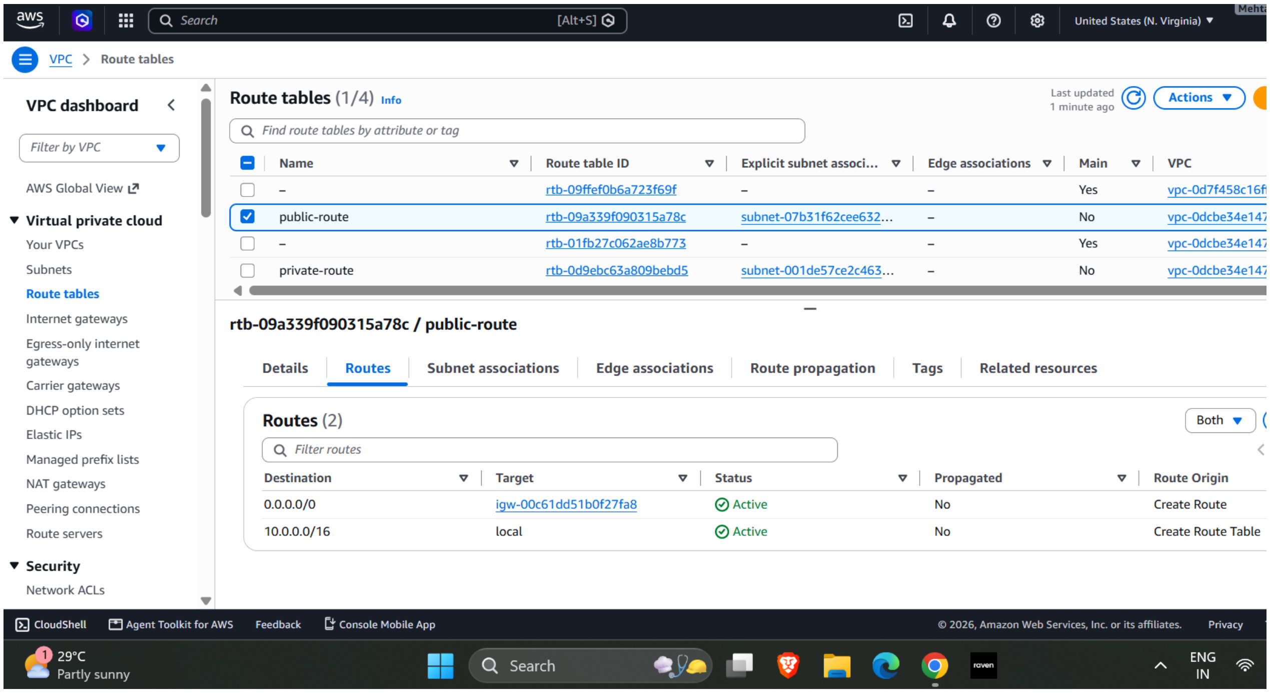
The private route table should normally contain only the local VPC route:

Associate Private Subnet

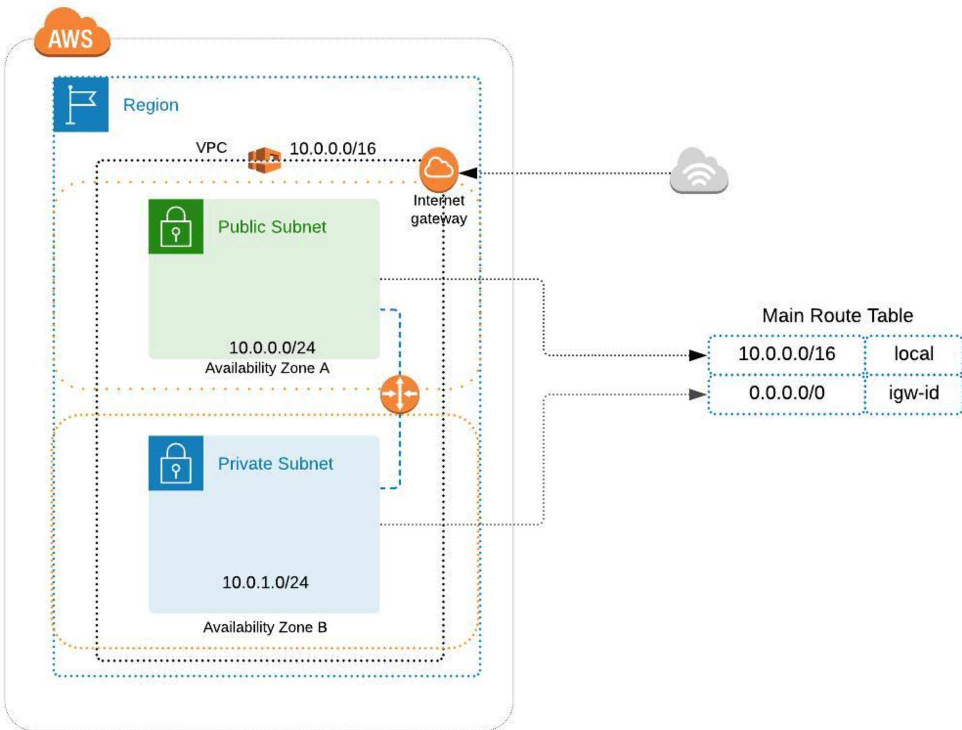
1. Select Private-Route
2. Go to **Subnet associations**.
3. Click **Edit subnet associations**.
4. Select:

Private-Subnet

5. Click **Save associations**.



5. Draw a simple diagram (hand-drawn or using any online tool) showing your VPC, public subnet, private subnet, and Internet Gateway, and briefly explain in 3-4 lines how this setup is similar to how apps like Zomato separate their public user interface from their private databases.



Internet → Internet Gateway → InstaVibeVPC

Inside the VPC:

- Public-Subnet (10.0.0.0/24)
- Private-Subnet (10.0.1.0/24)

This VPC setup is similar to how an app like Zomato can separate its public-facing services from its private backend systems. The public subnet can contain web servers that receive requests from users through the Internet Gateway. The private subnet can contain databases and backend services that should not be directly accessible from the internet. This separation improves security by limiting direct internet access to sensitive backend resources.